

Class 9 Biology – Tissues (Annual Examination 2026)

1. Define tissue. Explain its significance in multicellular organisms.

Answer: A tissue is a group of similar cells having a common origin, structure, and function. In multicellular organisms, tissues help in the division of labour, where different tissues perform specific functions efficiently. This increases the overall efficiency and survival of the organism.

Expected Keywords: Tissue, group of cells, common origin, division of labour, efficiency

2. What are meristematic tissues? State their characteristics. Differentiate between apical, lateral, and intercalary meristems.

Answer: Meristematic tissues are made of actively dividing cells responsible for growth in plants. They have thin cell walls, dense cytoplasm, large nucleus, and lack vacuoles. - **Apical meristem:** Located at root and shoot tips; increases length. - **Lateral meristem:** Found on sides of stems; increases girth. - **Intercalary meristem:** Present at nodes or leaf bases; helps in regrowth.

Expected Keywords: Actively dividing, growth, apical, lateral, intercalary

3. Differentiate between Parenchyma, Collenchyma, and Sclerenchyma.

Answer: - **Parenchyma:** Thin-walled, living cells; storage and photosynthesis. - **Collenchyma:** Unevenly thickened corners; provides flexibility. - **Sclerenchyma:** Thick lignified walls; provides strength and rigidity.

Expected Keywords: Thin-walled, thickened corners, lignin, support

4. What are aerenchyma and chlorenchyma? Where are they found?

Answer: Aerenchyma has large air spaces and is found in aquatic plants to help in buoyancy. Chlorenchyma contains chloroplasts and is found in green parts of plants for photosynthesis.

Expected Keywords: Air spaces, buoyancy, chloroplasts, photosynthesis

5. Why are coconut husks hard? Name the tissue responsible.

Answer: Coconut husks are hard due to the presence of sclerenchyma tissue with thick, lignified cell walls that provide mechanical strength.

Expected Keywords: Sclerenchyma, lignin, mechanical strength

6. Describe the role of epidermis in plants. How does it differ in desert plants?

Answer: The epidermis protects plants from water loss, injury, and pathogens. In desert plants, the epidermis has a thick waxy cuticle to reduce water loss.

Expected Keywords: Protection, cuticle, water loss

7. How is cork formed? Name the chemical present in its walls.

Answer: Cork is formed by secondary meristem called cork cambium. Its walls contain suberin, which makes it waterproof and protective.

Expected Keywords: Cork cambium, suberin, waterproof

8. Differentiate between xylem and phloem. List their components.

Answer: - **Xylem:** Conducts water; components are tracheids, vessels, xylem fibres, xylem parenchyma. - **Phloem:** Conducts food; components are sieve tubes, companion cells, phloem fibres, phloem parenchyma.

Expected Keywords: Water transport, food transport, tracheids, sieve tubes

9. Why is transport in xylem unidirectional and in phloem bidirectional?

Answer: Xylem transports water upward from roots to leaves, whereas phloem transports food in both directions depending on need.

Expected Keywords: Upward movement, source to sink, bidirectional

10. Name types of epithelial tissues and their locations.

Answer: - Squamous: Alveoli - Cuboidal: Kidney tubules - Columnar: Intestine - Stratified: Skin

Expected Keywords: Protection, absorption, lining

11. Which epithelium lines kidney tubules and respiratory tract? Role of cilia.

Answer: Cuboidal epithelium lines kidney tubules. Ciliated columnar epithelium in respiratory tract helps move mucus and dust particles.

Expected Keywords: Cuboidal, cilia, mucus movement

12. Why is blood called a connective tissue? Name plasma components.

Answer: Blood connects body parts by transporting substances. Plasma contains water, proteins, hormones, and salts.

Expected Keywords: Transport, plasma, proteins

13. Compare bone and cartilage.

Answer: Bone has a hard calcium-rich matrix, while cartilage has a flexible matrix and provides support.

Expected Keywords: Calcium, flexibility, support

14. Differentiate between ligament and tendon.

Answer: Ligaments connect bones to bones; tendons connect muscles to bones. Overstretching ligaments causes sprain.

Expected Keywords: Bone to bone, muscle to bone, sprain

15. State functions of areolar and adipose tissue.

Answer: Areolar tissue supports organs. Adipose tissue stores fat under the skin and around organs.

Expected Keywords: Support, fat storage, insulation

16. Compare striated, unstriated, and cardiac muscles.

Answer: Striated muscles are voluntary and multinucleate. Unstriated are involuntary. Cardiac muscles are involuntary and branched.

Expected Keywords: Voluntary, involuntary, branched

17. What is the unique feature of cardiac muscle?

Answer: Cardiac muscles have rhythmic contraction and do not get fatigued due to rich blood supply.

Expected Keywords: Rhythmic, non-fatiguing

18. Draw and label a neuron. State function of axon and dendrites.

Answer: Dendrites receive impulses; axon transmits impulses away from the cell body.

Expected Keywords: Impulse, dendrites, axon

19. List three differences between plant and animal tissues.

Answer: Plant tissues are mostly dead, rigid, and show continuous growth. Animal tissues are living, flexible, and show limited growth.

Expected Keywords: Rigid, flexible, growth

20. Which tissue helps in repair of injured tissues?

Answer: Areolar connective tissue helps in healing and filling spaces inside organs.

Expected Keywords: Repair, healing, areolar tissue